

Metamaterials for Water Waves: Theory, Design, and Applications

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Metamaterials, initially developed for electromagnetic and acoustic wave manipulation, have recently emerged as a powerful tool for controlling water waves. These engineered structures exhibit unique properties not found in natural materials, enabling unprecedented control over wave propagation, refraction, and focusing. This review offers a comprehensive overview of metamaterials for water wave applications, emphasizing their theoretical foundations, design methodologies, and practical implementations. It begins by outlining the fundamental physics of water waves and the key equations governing wave phenomena. Then, the principles and design of metastructures, which serve as the fundamental building blocks of water wave metamaterials, are explored. This focus includes various metadevices, such as invisibility cloaks, wave concentrators, rotators, and superscatterers. Through an extensive analysis of both numerical and experimental studies, this paper provides a roadmap for future research and technological advancements in water wave metamaterials, highlighting their potential to address challenges, particularly in ocean energy harvesting, coastal defense, and disaster prevention.

1. Introduction

The manipulation of water surface waves has emerged as a transformative approach with wide-ranging implications for ocean energy harvesting, coastal defense, and disaster prevention. Traditional approaches face significant challenges due to the complexity of natural water environments, where factors like wind, terrain, and varying water depths complicate wave behavior. Metamaterials—engineered structures with tailored properties—offer an alternative approach by enabling control over key parameters such as frequency, amplitude, and wavelength.^[1–3] In this context, water wave metamaterials provide new opportunities for enhancing coastal management and improving the efficiency of wave energy conversion.

The broader concept of artificial composite materials has been explored for more than a century, with progress driven by advances in design and fabrication. Early studies of wave propagation in periodic structures laid the foundation, while the

emergence of photonic crystals initiated modern wave manipulation strategies. The introduction of metamaterials demonstrated that subwavelength building blocks can collectively mimic continuous materials while supporting exotic wave behaviors absent in natural systems. These developments led to breakthroughs across multiple domains, including optics,^[4] acoustics,^[5] electromagnetics,^[6–8] thermodynamics,^[9] and water waves.^[10–20] Their ability to manipulate waves at subwavelength scales has already enabled phenomena such as invisibility cloaking,^[21–26] analogue computing,^[27,28] quantum photonics,^[29,30] and wireless communication.^[31–33]

By exploiting analogies between Maxwell's equations and water wave equations, many of these concepts have been successfully extended to hydrodynamics. Water wave metamaterials have since been used to demonstrate superlensing,^[34] negative effective gravity,^[35–37] focusing,^[38–40] topological effects,^[41] Bloch waves,^[42–44] and unidirectional transmission.^[45,46] Their design relies on determining equivalent material parameters through methods such as homogenization, transformation optics, and scattering cancellation. Importantly, gradient metamaterials—with spatially varying refractive indices—allow fine control of wave trajectories, enabling functionalities not achievable with uniform structures.

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Recent experiments have confirmed the feasibility of metadevices such as water wave cloaks,^[47–59] concentrators,^[60–63] rotators,^[64] shifters,^[11] superscatterers,^[65] and absorbers.^[66–69] For example, cylindrical cloaks can bend surface waves and isolate internal fields,^[70] while concentrators intensify localized wave energy and rotators redirect internal fields to a desired orientation. These results highlight the growing potential of metamaterial-based approaches in hydrodynamic systems.

In this Review, we summarize recent progress in water wave metamaterials and metadevices. We first introduce the fundamental physics of water waves and the governing equations. Next, we outline the design principles of metastructures that act as the building blocks for metamaterials. Finally, we examine key metadevices—including cloaks, concentrators, rotators, and superscatterers—and discuss their applications. We conclude by identifying the current challenges and future directions, with particular emphasis on the transition from proof-of-concept demonstrations to scalable, real-world implementations.

2. Principle of Water Waves and Metamaterials Design

2.1. The Physics of Water Waves

Water waves are mechanical waves traveling across the water surface, with gravity acting as the primary restoring force.^[71] Because they propagate at the air–water interface, they are commonly referred to as surface gravity waves. The close analogy between electromagnetic (EM) waves and water waves makes it possible to extend principles of EM scattering theory to hydrodynamic systems. In particular, the governing equation of water waves closely resembles that of EM waves, especially under shallow-water conditions. In this regime, the effective refractive index of water waves can be tuned simply by altering the relative water depth.^[72] This establishes a direct pathway for applying metamaterial concepts to the control of water waves.

In the study of water-wave metamaterials, the starting point is the description of linearized wave motion under classical hydrodynamic assumptions. When the fluid is treated as inviscid, incompressible, and irrotational, the velocity potential $\psi(x, y, z)e^{-i\omega t}$ satisfies the three-dimensional Laplace equation:

Under the assumptions that the fluid is inviscid, incompressible and irrotational, the propagation of harmonic, linearized water waves is governed by the 3D Laplace's equation

$$\nabla^2 \psi = 0 \quad (1)$$

subject to appropriate boundary conditions. At the free surface ($z = 0$), the kinematic and dynamic constraints reduce to

$$\frac{\partial}{\partial z} \psi = \frac{\omega^2}{g} \psi \quad (2)$$

where ω is the angular frequency and g denotes gravitational acceleration. At the surfaces of rigid, submerged bodies, I , the impermeability condition requires that

$$\frac{\partial}{\partial n} \psi = 0 \quad (3)$$

where n is the unit vector normal to surface I . These relations form the fundamental framework for water-wave control, valid in the limit of linear, small-amplitude waves where the wave height is much smaller than both the wavelength λ and water depth h . Because the governing equations describe a fully three-dimensional system, direct numerical solutions are often computationally demanding. This has motivated the development of effective-medium approaches and homogenization methods to approximate water-wave behavior in structured environments, enabling the practical design of metamaterial devices.

When the variation in depth is small compared with the wavelength ($kd \ll 1$), linear water wave theory can be applied. In this case, the mild-slope equation (MSE) provides a convenient description of the surface elevation, forming the theoretical foundation for subsequent studies on water wave metamaterials:

$$\nabla \cdot (d \nabla \xi) + \frac{\omega^2}{g} \xi = 0 \quad (4)$$

where the free-surface elevation is represented by ξ , the local water depth is d , the frequency of the incident plane wave is ω , and the gravitational acceleration is g . The mild-slope equation describes the combined refraction-diffraction phenomenon for simple harmonic waves. The linear dispersion relation for water waves is given as:

$$\omega^2 = gk \tanh(kd) \quad (5)$$

where the local water wave number is denoted as k . The dispersion relation for the shallow water approximation can be expressed as:

$$\omega^2 = gk^2 d \quad (6)$$

for the case where $kd \gg 1$, indicating that the wavelength is much shorter than the water depth, Equation (4) can be expressed as:

$$\nabla^2 \xi + \frac{\omega^4}{g^2} \xi = 0 \quad (7)$$

and the corresponding dispersion relation for deep water is:

$$\omega^2 = gk \quad (8)$$

This implies that the dispersion relation is independent of depth in deep water regions.

2.2. Design Principles for Water-Wave Metamaterials

The design of water-wave metamaterials centers on deriving effective medium parameters that translate structural geometry into macroscopic control of wave propagation. A key step is homogenization, where periodic or quasi-periodic scatterers are represented by equivalent depth and effective gravitational acceleration. This requires solving the boundary-value problem for wave fields inside and outside the structure, enforcing continuity of surface elevation and flow. Under zero-scattering conditions, the homogenized parameters can be expressed in terms of structural characteristics such as filling fraction, geometry, and lattice

Table 1. Key structural parameters of water-wave metamaterials.

Parameter	Symbol/definition	Typical range or control method	Primary influence on metamaterial behavior
Scatterer radius	r	Typically $0.1\text{--}0.5a$	Controls scattering strength and local phase delay, influencing effective refractive index contrast
Unit-cell/lattice spacing	a	Typically $a \ll \lambda$ for homogenized designs	Determines the validity of the effective medium approximation and periodicity-driven dispersion
Filling fraction	$f = \frac{A_{\text{scatterer}}}{A_{\text{cell}}}$	Adjusted via geometry or packing ratio	Sets the dispersion relation through $\tan h(kh)$, affecting wave speed and resonance conditions
Water depth	h	Laboratory: $5\text{--}200$ mm; Coastal: $0.5\text{--}50$ m	Modifies the dispersion relation and the effective gravitational acceleration parameter
Depth gradient/bathymetry profile	$h(x,y)$	Continuous or stepwise modulation	Enables graded-index focusing, wave redirection, and cloaking
Porosity/permeability	p	Controlled via perforation or porous materials	Introduces tunable dissipation and leakage, relevant to amplitude shaping and breakwater applications
Lattice symmetry	–	Square, hexagonal, quasi-periodic	Determines isotropy, band gaps, and directional selectivity
Number of layers (effective thickness)	t	Typically $1\text{--}10$ or continuous bathymetry	Affects resonance, transmission-reflection balance, and bandwidth of controllable wave modes
Surface/profile modulation	–	Patterned, stepped, or continuous shaping	Enables multifunctionality, such as focusing, combined with spectral filtering

arrangement. For canonical designs like cylindrical arrays, these parameters can be calculated directly from the scatterer-to-cell area ratio.

For more advanced architectures, such as stepped- or gradient-index metamaterials, spatial refractive index profiles are engineered by adjusting cylinder radii, density, or local water depth. This approach enables diverse wave manipulations—including focusing, beam steering, and cloaking—analogueous to transformation optics in electromagnetics. For instance, a hyperbolic secant profile realized through gradual radius variation has been used to construct water-wave lenses with enhanced focusing capability.

At longer wavelengths, when the structural period is much smaller than the wavelength, effective medium theory (EMT) provides a reliable approximation for homogenized parameters. EMT simplifies broadband design, predicts dispersive behavior, and supports the realization of anisotropic metamaterials through asymmetric unit cells or layered seabed structures. Collectively, these theoretical tools bridge fundamental wave physics with practical metadvice implementation.

The theoretical framework discussed above establishes the correspondence between structural geometry and effective medium parameters. The following subsection enumerates the key geometric and material descriptors that govern this structure–function relationship.

2.3. Structural Parameters

The effective behavior of water-wave metamaterials is governed not only by the underlying hydrodynamic dispersion relation but also by a set of key structural parameters that determine scattering strength, refractive properties, anisotropy, and dissipation characteristics. Similar to mechanical and acoustic metamaterials, the geometric configuration and material attributes of the unit cell play central roles in shaping the macroscopic wave response. **Table 1** summarizes the primary parameters relevant to

water-wave metamaterial design, including scatterer geometry, lattice periodicity, filling fraction, and bathymetric profile, as well as porosity, structural symmetry, and material rigidity. Clarifying the influence of these parameters provides a practical foundation for design optimization, performance tuning, and the systematic comparison of different device architectures across laboratory and coastal-scale implementations.

2.4. Building Blocks of Water Wave Metamaterials

Water wave metamaterials are typically composed of periodic subwavelength structures that derive novel effective properties from their geometry. Common designs include submerged pillars, seabed perforations, and controlled variations in depth. The analysis of such systems often relies on homogenization techniques, which replace heterogeneous microstructures with equivalent continua for easier modeling.^[73–76] For instance, arrays of thin plates have been proposed to realize effective metamaterials for water waves.^[77,78] By applying homogenization, the complex scattering of individual elements can be reduced to tractable effective parameters. While this approach assumes subwavelength periodicity, exotic wave behaviors may also arise when the structural scale approaches the wavelength.^[79,80]

To achieve tailored functionalities, it is essential to determine the effective material parameters of these metamaterials. For circular or cylindrical geometries, methods such as the coherent potential approximation can be employed, with parameters derived from filling fractions. In cylindrical systems, amplitudes inside and outside the effective medium can be obtained in a cylindrical coordinate framework, ensuring continuity and zero scattering at the boundaries. When the wavelength is much larger than the lattice period, effective medium theory provides a reliable description. A representative example is a two-dimensional periodic lattice of rigid cylinders, which can realize superlensing effects for surface waves (**Figure 1a**).^[34] Such structures enable

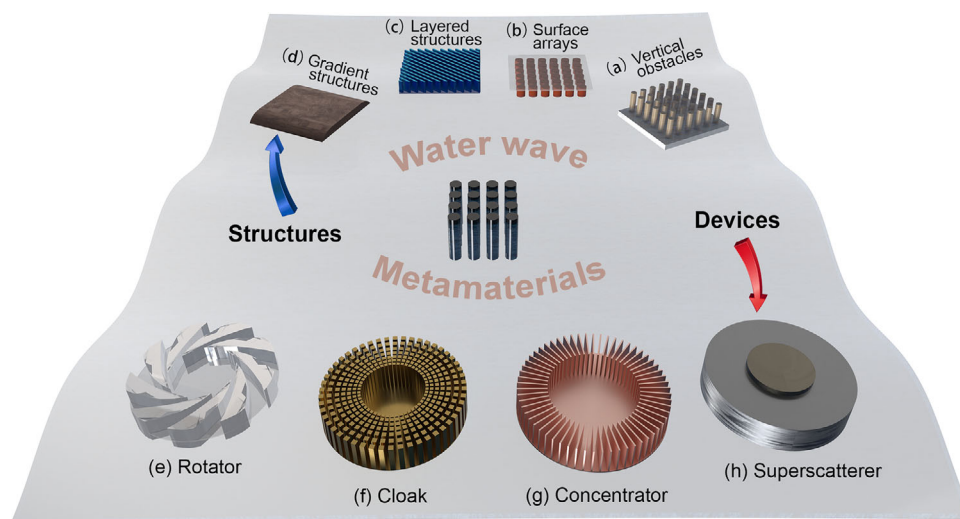


Figure 1. Representative structures for water wave manipulation (a–d) and metamaterials (e–h). a) Vertical obstacle structures. b) Surface arrays. c) Layered structures. d) Gradient structures. e) Water Wave rotator for controlled redirection of incident wave energy. f) Cloaking structures designed to minimize scattering and achieve wave invisibility. g) Wave concentrator for energy localization and amplification. h) Superscatterers that enhance scattering cross-sections beyond conventional limits. Panels a–d illustrate metamaterial building blocks, which are engineered media with subwavelength characteristics. Panels e–h depict metamaterials, which are application-oriented implementations of metamaterials designed for targeted wave-manipulation tasks, including cloaking, beam steering, and wave focusing. This distinction is maintained throughout the manuscript.

controlled refraction of water waves and have been demonstrated in water-wave focusing lenses. Another approach is the use of vertically oriented, bottom-mounted split tubes arranged in a periodic array. This design yields an effective negative gravity, providing a means to block incident water waves. These concepts not only advance the fundamental understanding of wave manipulation but also suggest practical applications in wave-energy harvesting and coastal protection.^[81]

Experimental demonstrations further highlight these principles. Periodic arrays of rigid disks on the water surface can reduce effective depth while enhancing gravitational acceleration.^[45] Similarly, fixed arrays of surface-piercing cylinders can create a complete band gap, preventing wave propagation in all directions (Figure 1b).^[82] Based on this principle, semicircular arrays of surface disks have been used to obstruct omnidirectional waves from point sources.

Beyond periodic lattices, other configurations extend the design space. Arrays of rigid plates forming subwavelength gratings have been proposed for wave redirection.^[83] Anisotropic metamaterials built from closely spaced vertical plates have been ex-

perimentally shown to suppress backscattering in sharply bent channels.^[84] Layered structures, as in Figure 1c, can guide waves along curved paths, while gradient index metamaterials enable precise refractive control.^[11]

Gradient water wave metamaterials differ from conventional designs by enabling spatially varying refractive index profiles. This is achieved by altering effective water depth or related parameters, allowing precise control of wave propagation. For instance, an array of bottom-mounted vertical cylinders can tailor the effective depth and gravity constant, forming the basis of water-wave focusing lenses.^[38] Building on this principle, a gradient-index lens has been realized by varying the cylinder radii to generate a hyperbolic secant index distribution, as illustrated in Figure 1d and Table 2.^[40]

3. Metamaterials for Water Wave Manipulation

Metamaterials provide powerful means to manipulate water waves in ways not achievable with natural media.^[85–88] Similar to advances in microwaves,^[89–95] optics,^[96] and acoustics,^[97] where

Table 2. Building blocks of water wave metamaterials and applications.

Metamaterial structures	Application	Refs.
Vertical obstacle structures	Superlensing effect in liquid surface waves	[34]
	Experimental observation of negative effective gravity	[81]
Surface arrays for tailored	Fast water waves in stationary surface disk arrays	[45]
	Achieving Complete Band Gaps of Water Waves	[82]
Layered structures	Rainbow reflection.	[83]
	Experimental realization of a water-wave metamaterial shifter	[11]
Gradient structures	Refraction of water waves	[38]
	Focusing on liquid surface waves	[40]

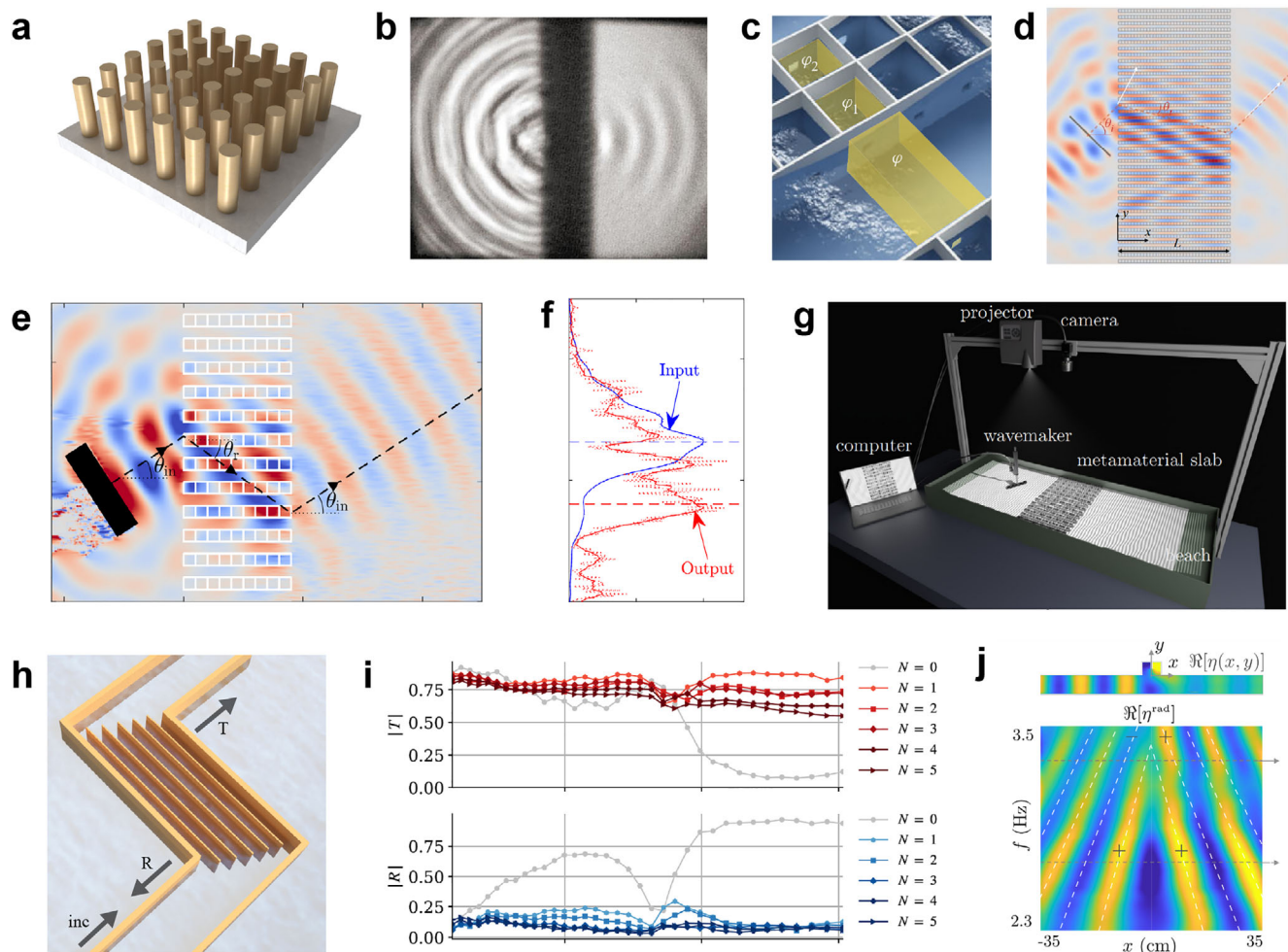


Figure 2. Negative refraction and absorption of water waves. a) Negative refraction of liquid surface waves using a fabricated metamaterial slab composed of a periodic array of copper cylinders on a rectangular base. b) Experimental projection of surface wave patterns onto a screen, where the central black strip corresponds to the copper cylinder slab. A real image is clearly formed behind the slab due to negative refraction at 6.15 Hz. Reproduced with permission.^[34] Copyright 2004, American Physical Society. c) Negative refraction achieved with hyperbolic metamaterials. Shown are the unit cells of the single- and double-resonant canal structures used to derive the effective propagation properties. d) Experimental results confirming negative refraction of water waves through the hyperbolic metamaterial slab. c,d) Reproduced with permission.^[106] Copyright 2023, The Author(s), Published by Cambridge University Press. e) Hyperbolic-dispersion metamaterials enabling negative refraction of water waves. Fourier transform profilometry (FTP) measurements of the surface elevation show an incident beam undergoing negative refraction within the metamaterial slab. f) Corresponding normalized wave amplitudes. g) Schematic of the experimental setup for the negative refraction measurement. e-g) Reproduced with permission.^[110] Copyright 2024, American Physical Society. h) Suppression of backscattering in sharply bent water wave channels. Schematic of the experimental setup. i) Measured reflection and transmission coefficients for varying numbers of plates. Reproduced with permission.^[84] Copyright 2023, American Physical Society. j) Experimental demonstration of perfect resonant absorption of guided water waves, highlighting the distinct monopolar and dipolar resonance modes. Reproduced with permission.^[125] Copyright 2023, American Physical Society.

metamaterials have transformed wave manipulation, engineered structures can also reshape the behavior of water waves. In this section, we review key mechanisms for water wave manipulation, including negative refraction, absorption, anisotropy, and unidirectional transmission. Together, these approaches establish the foundation for advanced control of surface wave phenomena.

3.1. Negative Refraction of Water Waves

Wave manipulation through negative refraction is one of the most studied topics in metamaterials. Pendry proposed that

negative refraction could enable a superlensing effect.^[98] This phenomenon has since been demonstrated for visible light,^[99] microwaves,^[100,101] acoustics,^[102–105] and water waves.^[34,106,107]

For water waves, a superlensing experiment was demonstrated by arranging surface-piercing copper cylinders into a metamaterial slab. This structure enabled negative refraction of liquid surface waves, resulting in the formation of a real image at 6.15 Hz (Figure 2a,b).^[34] More recently, anomalous dispersion was observed with inclined plates, which directed energy flow unidirectionally.^[107] In addition, subwavelength resonant media have been engineered to induce either elliptic or hyperbolic dispersion, thereby enabling negative refraction of water waves.^[106]

This effect has been demonstrated using hyperbolic metamaterials, where unit cells based on single- and double-resonant canal structures were designed to tailor the effective propagation properties (Figure 2c). Experimental observations further confirmed negative refraction through the metamaterial slab, validating the predicted hyperbolic dispersion behavior (Figure 2d). These designs allow wave propagation along interfaces without backscattering, even in the presence of defects.

A particularly promising strategy relies on hyperbolic dispersion.^[108,109] Locally resonant metamaterials have been fabricated to exhibit both elliptic and hyperbolic regimes, with negative refraction realized in the latter.^[110] This phenomenon has been experimentally validated using Fourier Transform Profilometry (FTP) measurements of surface elevation, which reveal an incident wave beam undergoing negative refraction within the hyperbolic-dispersion metamaterial slab (Figure 2e), along with the corresponding normalized wave amplitudes (Figure 2f) and the experimental setup (Figure 2g).

Although most studies address small-amplitude linear waves, nonlinear water wave metamaterials remain an important open frontier.^[111–115]

3.2. Absorption of Water Waves

Absorption is critical for applications such as energy harvesting and noise reduction. Achieving perfect absorption in compact devices has attracted wide interest across optics,^[116,117] microwaves,^[118–120] acoustics,^[121,122] and water wave.^[84,123–126]

In water waves, the motivation is often to minimize reflections in harbors and basins. For example, two closely spaced resonant channels have been designed as scatterers in a waveguide, enhancing absorption. A plate-array metamaterial composed of vertically oriented thin plates has also been demonstrated to reduce backscattering in sharply bent channels, with the experimental setup shown schematically (Figure 2h) and the corresponding reflection and transmission measurements for different numbers of plates presented in Figure 2i.^[84]

Perfect absorption has been pursued using Autler–Townes splitting resonance, experimentally demonstrated through monopolar and dipolar resonances of guided water waves (Figure 2j).^[125] An asymmetric point scatterer with dual resonant side channels was tuned to balance energy leakage and viscous losses. Other strategies include arrays of periodic floating oscillators for enhanced attenuation^[127] and submerged resonators that leverage both local resonance and Bragg scattering to achieve broadband absorption.^[128] These approaches show promise for coastal protection and wave attenuation.

3.3. Anisotropy of Water Waves

Many metamaterial designs rely on extreme anisotropy, which is challenging to achieve. Subwavelength bathymetry offers a solution, allowing waves to experience different effective depths depending on propagation direction.^[16,129]

Water wave metamaterials can display stronger anisotropy than their electromagnetic or acoustic counterparts. This arises from the shallow-water approximation, which neglects three-dimensional near-field effects. The anisotropic response of water

waves is experimentally demonstrated using a metamaterial realized through a subwavelength-layered variation of bathymetry. A slender tip is employed to excite waves propagating across the structure, and space- and time-resolved measurements of the free surface elevation are obtained via Fourier Transform Profilometry at 2 and 3 Hz (Figure 3a,b).^[130] Furthermore, experiments with arrays of rigid surface disks showed increased effective gravity and reduced water depth, resulting in higher phase and group velocities.^[45] Such designs enable total reflection at oblique incidence and unidirectional transmission.

Transformation-based strategies extend these concepts. A circular wavelength modulator was proposed using spatial transformation to compress and expand the wave domain, acting as both a wavelength reducer and amplifier.^[131] Experimental and numerical investigations of metabathymetry in deformed cavities demonstrate that carefully designed depth profiles can faithfully replicate the resonant modes observed in standard rectangular cavities. This capability allows for precise control of sloshing dynamics even in irregular geometries, effectively enabling the manipulation of wave patterns and energy distribution within the cavity (Figure 3c–e).^[132]

3.4. Unidirectional Water Wave Transmission

Controlling the directionality of wave propagation is vital for offshore and coastal applications. Unidirectional transmission improves the efficiency of wave energy converters and enhances coastal protection.^[133,134] Several approaches have been reported. A one-dimensional immersed system achieved unidirectionality through dispersion–diffraction interplay.^[135] Channels with periodic sidewalls have been used under deep-water conditions to guide waves asymmetrically.^[136] Inspired by the ancient Luoyang Bridge, a metagrating was constructed to regulate surface waves and achieve unidirectional propagation.^[137]

Analogies to spoof surface plasmon polaritons in electromagnetics^[138–142] motivated a rotating screw-propeller structure for water waves, creating unidirectional surface polaritons. Unidirectional water waves are achieved using a one-dimensional groove array, which selectively guides wave propagation along a preferred direction (Figure 3f).^[46] Numerical simulations capture the resulting field pattern, revealing the effectiveness of the groove design in directing energy flow (Figure 3g). Experimental measurements further confirm the unidirectional behavior, with the setup illustrated in Figure 3h, demonstrating precise control over water wave directionality. Another emerging direction is Willis coupling, originally studied in acoustics.^[143–147] A crescent-shaped pedal supported by a mushroom-shaped pillar was proposed as a Willis metamaterial for water waves, enabling distinct impedances for forward and backward propagation (Figure 3i). Wave shoaling above each pillar, caused by an abrupt decrease in water depth, reduces the phase velocity and induces resonance, further enhancing the asymmetric wave response (Figure 3j).

4. Metadevices for Water Waves

Metamaterials, with their unique ability to manipulate wave propagation, have opened new possibilities for controlling water waves in innovative ways. In this section, we explore various

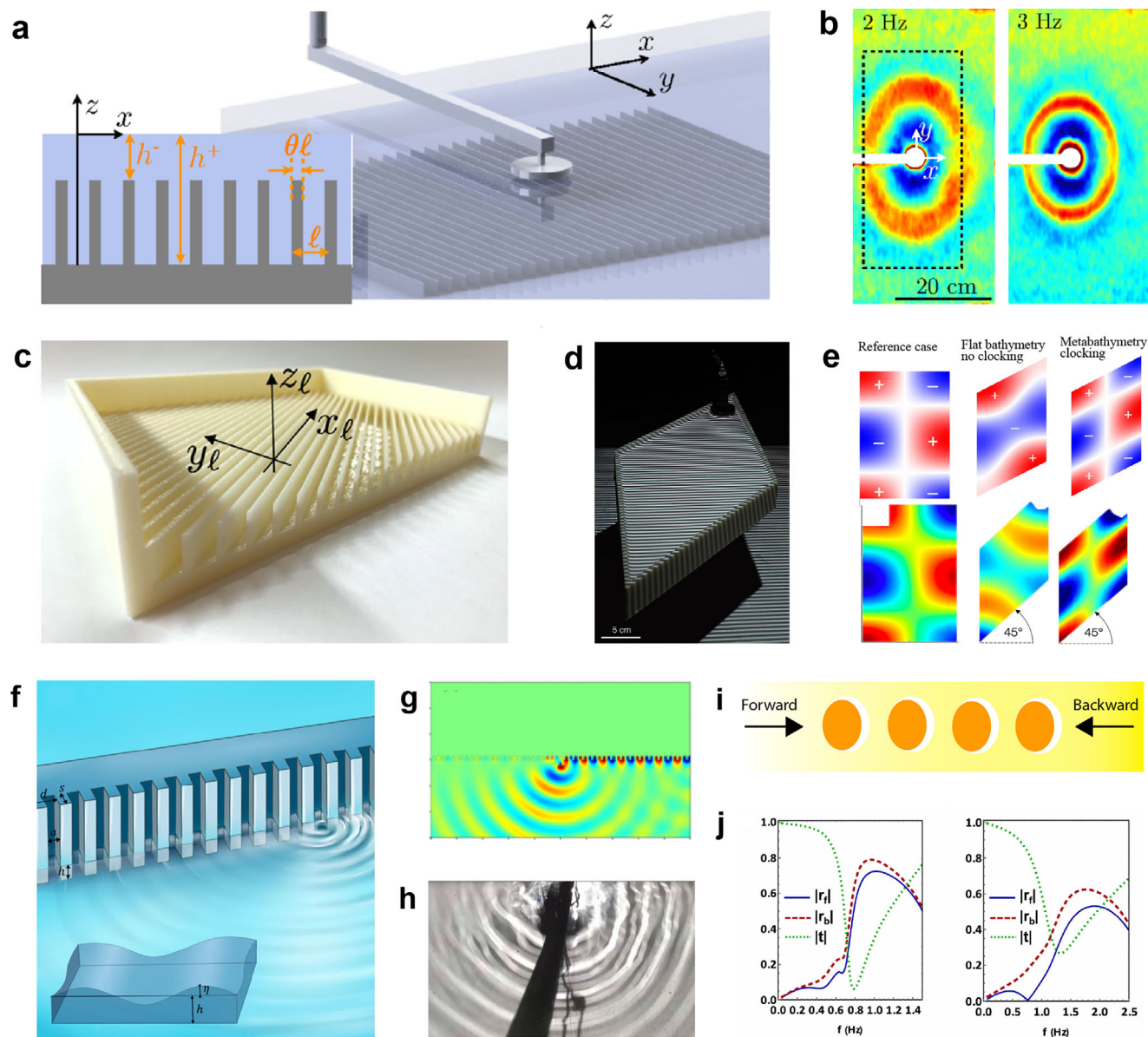


Figure 3. Anisotropy and Unidirectional transmission of water waves. a) Experimental demonstration of water wave anisotropy. The metamaterial is realized through a subwavelength-layered variation of bathymetry, and a slender tip is used to excite water waves propagating across the structure. b) Space- and time-resolved measurements of the free surface elevation are obtained via Fourier Transform Profilometry, enabling precise characterization at 2 and 3 Hz. a,b) Reproduced with permission.^[130] Copyright 2017, American Physical Society. c) Regular sloshing modes observed in irregular cavities. The designed metamaterial cavity sample is shown. d) Experimental setup for the measurements. e) Comparison of a rectangular reference cavity, a flat bathymetry without cloaking, and a metabathymetry-enabled cavity with cloaking. c-e) Reproduced with permission.^[132] Copyright 2024, American Institute of Physics. f) Schematic of the generation of unidirectional water waves. g) Simulated field pattern of water waves excited by a one-dimensional groove array. h) Experimental setup used to validate the unidirectional wave propagation. f-h) Reproduced with permission.^[146] Copyright 2022, American Physical Society. i) Willis coupling in water waves. The unit cell consists of a mushroom-shaped pillar topped with a crescent-shaped wall. j) Wave shoaling above each pillar occurs due to an abrupt decrease in water depth, reducing the phase velocity and inducing resonance. i,j) Reproduced with permission.^[147] Copyright 2021, The Institute of Physics.

metadevices such as cloaks, concentrators, rotators, shifters, and superscatterers that effectively control water wave behavior. We also examine the design of superscatterers, wave shifters, and rotators, highlighting their potential for diverse practical uses.

These advancements demonstrate the versatility of metamaterials in enhancing water wave management and expanding their potential applications in ocean energy harvesting and coastal protection.

4.1. Water Wave Cloaks

The concept of invisibility cloaks, long imagined in science and fiction, has become tangible through metamaterials. Initially developed in optics,^[148–150] the principle has since been extended to microwaves,^[151–155] acoustics,^[156] elastic waves,^[157,158] thermal fields,^[159] and water waves.^[70,160–162] For water waves, cloaking aims to suppress scattering so that objects become nearly undetectable to incident waves. The historical development of these water wave cloaks is summarized in a timeline (Figure 4a).

4.1.1. Cylindrical Cloaks

One of the earliest demonstrations involved a cylindrical water-wave cloak designed to minimize scattering.^[70] The structure consisted of 100 angular sections with seven rows of rods along the radius (Figure 4b). Using homogenization theory, adjacent cylinders were shown to generate anisotropic shear viscosities that bend waves smoothly around a central region. The effectiveness of this design was confirmed by diffraction measurements, where a rigid cylinder enclosed by the structured cloak exhibited significantly reduced scattering (Figure 4c) compared with the strong diffraction observed for an uncloaked rigid cylinder (Figure 4d). This approach illustrates the potential of structured cloaks but also highlights the complexity of the required material architectures.

4.1.2. Bathymetry-Based Cloaks

For shallow-water waves, transformation-optics-based cloaking requires spatial variations in two independent material properties, typically permittivity and permeability. When extended to gravity waves, the analogous parameters become gravitational acceleration and water depth. However, because gravitational acceleration cannot be altered in practice, this constraint limits direct implementation of conventional transformation-based cloaking approaches for water waves. To circumvent this limitation, a theoretical broadband cloak has been proposed for stratified seas, designed to shield floating objects by tailoring bottom corrugations. This method exploits the nonlinear resonance between surface waves, interfacial waves, and seabed topography—a phenomenon arising from the dispersive nature of gravity waves—thus offering a nonlinear transformation alternative to transformation optics for dispersive wave systems.^[49]

The central advantage of bathymetry-based cloaking lies in its reliance solely on varying water depth while keeping gravitational acceleration constant. The nonlinear transformation preserves the governing equations and, when the cloak is sufficiently large, asymptotically satisfies the boundary conditions required for invisibility.^[52] Numerical studies demonstrate that by modulating the seabed profile around a vertical circular cylinder, the scattered wave energy can be almost entirely suppressed.^[51] As a result, the object becomes effectively invisible to distant observers, highlighting the potential of bathymetry as a practical route to broadband water-wave cloaks.

Further investigations reveal that when a plane wave impinges on a surface-piercing circular cylinder inside a waveguide,

strong Fano resonances emerge, leading to intense scattering. Such resonances pose a challenge for cloaking efficiency. However, carefully optimized bathymetric designs can suppress both resonant and background scattering across a broad frequency range, thereby enabling reflectionless cloaking. This effect has been confirmed through numerical simulations and experimental validations, demonstrating the robustness of bathymetry-based cloaks for controlling water-wave scattering.^[163]

4.1.3. Scattering Cancellation Cloaks

As water depth increases, the influence of bottom topography diminishes exponentially, making seabed modifications ineffective for wave control. Additionally, the governing equation for deep-water waves under potential flow is not invariant under coordinate transformations, complicating cloaking techniques. To address this, scattering cancellation methods have been proposed for finite depths, where seabed topography can be designed to eliminate scattered wave energy around a bottom-mounted cylinder.^[164] Utilizing multiple floating cylinders or rings has also shown effective cloaking performance for deep-water waves. A numerical study has demonstrated water wave cloaking for a truncated cylinder surrounded by several additional cylinders.^[57] Furthermore, a periodic quasi-cloaking effect has been reported by adjusting the spacing between inner and outer cylinders.^[165] Cloaking based on scattering cancellation has been validated through numerical computations using a higher-order boundary element method^[166] and model experiments.^[167] When using an asymmetrical array of cylinders, the cloak's effectiveness is wave direction-dependent.^[168] A cloaking method based on scattering cancellation involves using a floating composite elastic plate that encircles the cylinder, effectively neutralizing the waves scattered by the cylinder.^[58]

4.1.4. Carpet Cloaks

While the theory behind transformation optics-based cloaking is conceptually sophisticated, its practical realization presents considerable challenges because of the required material compositions with extreme parameters. For instance, the design of a hydrodynamic cloak requires the anisotropic distribution of water depth and variations in gravitational acceleration, posing significant obstacles to practical implementation. Often, relaxing these constraints leads to impedance mismatches and reduced performance. Furthermore, cloaks designed through transformation optics are typically narrowband and prone to losses due to specific requirements for local superluminal parameters. Consequently, alternative methods for manipulating electromagnetic waves, such as metasurfaces, have garnered significant research interest.^[169–181] By extending transformation optics to water waves, a carpet cloak has been developed by arranging alternating periodic water depths. The experimental setup conceals a bump beneath the cloak (Figure 4e), and both experimental measurements (Figure 4f) and numerical simulations (Figure 4g) confirm the excellent invisibility performance of this design.^[56] An invisibility carpet has been proposed to manage surface wave reflection in a bounded wave channel, validated through both

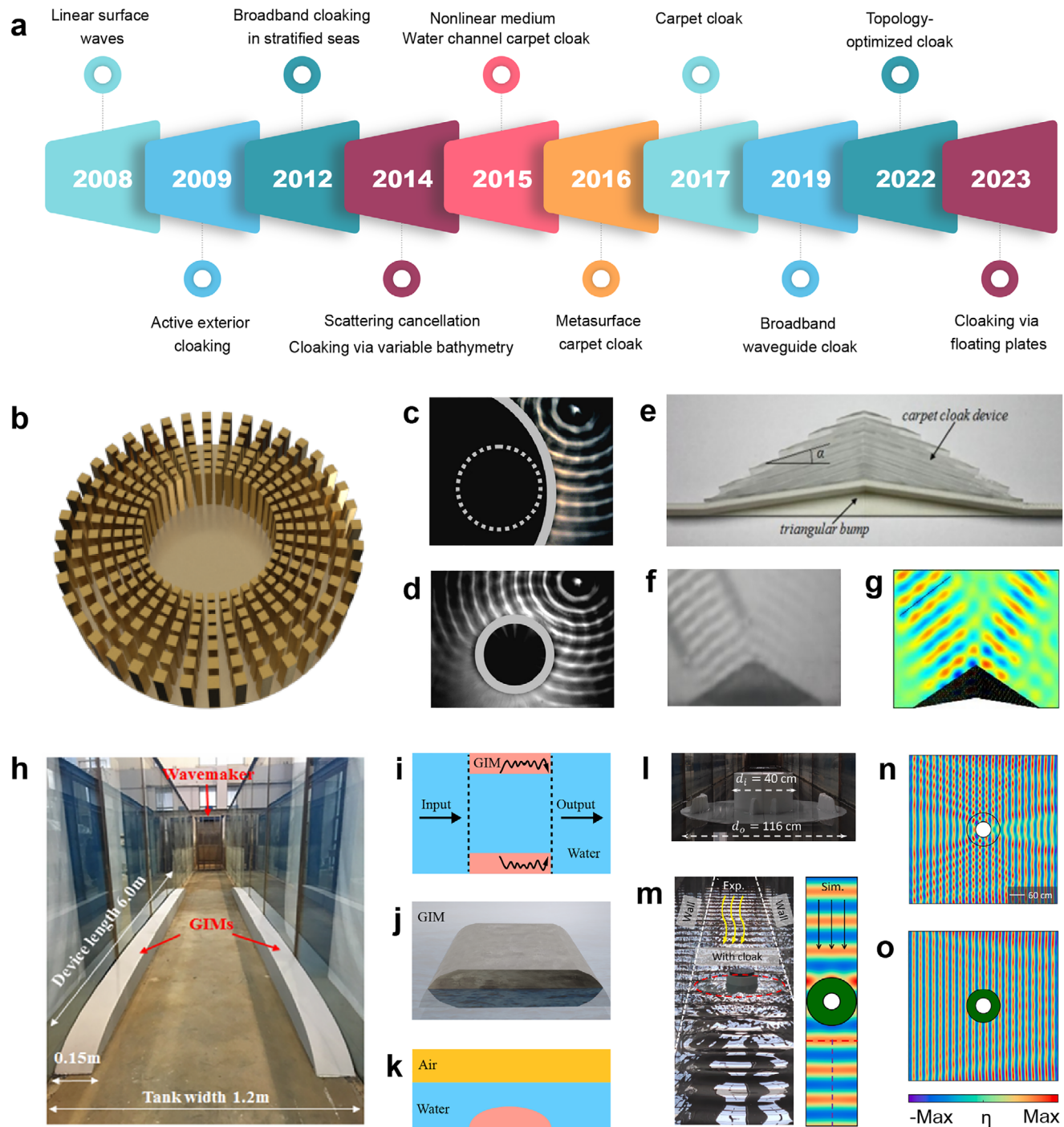
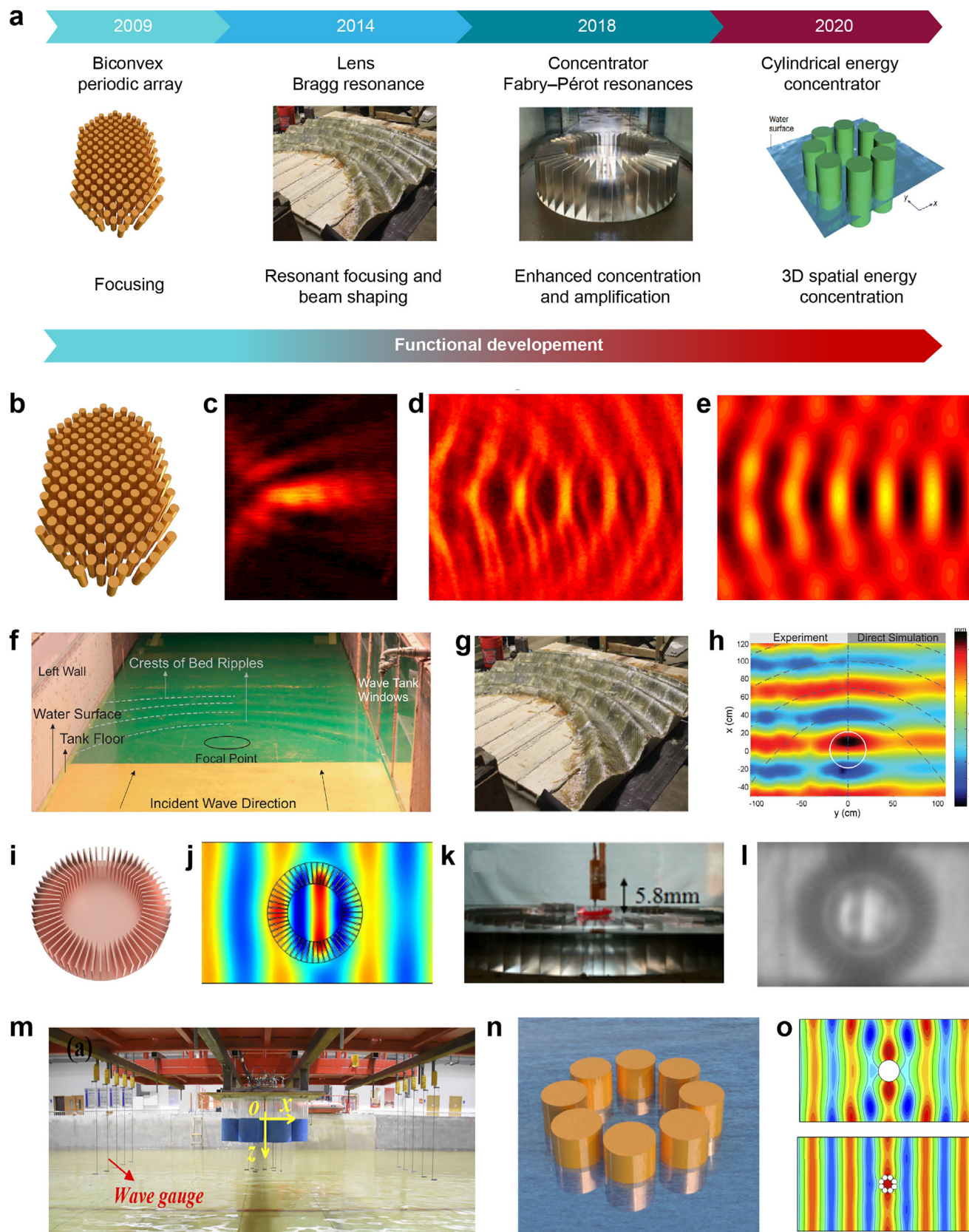


Figure 4. Water wave cloaks. a) Timeline of water wave cloaks. b) Fabricated sample of the annular structured cloak. c) Diffraction pattern of surface waves around a rigid cylinder enclosed by the structured cloak. d) Diffraction pattern of surface waves around a rigid cylinder without the cloak. c,d) Reproduced with permission.^[70] Copyright 2008, American Physical Society. e) Carpet cloak: experimental setup with a surface bump concealed beneath the cloak. f) Experimental results demonstrating the cloaking effect. g) Numerical simulations validating the experimental observations. e-g) Reproduced with permission.^[56] Copyright 2017, American Physical Society. h) Experimental setup of a broadband cloak for water waves. i) Top view of the cloaking device. j) Structural design of the gradient-index metamaterial (GIM). k) Side view of the cloak incorporating the GIM structure. h) Reproduced with permission.^[161] Copyright 2019, American Physical Society. l) Experimental photograph of the topologically optimized cloak. m) Schematic illustration of the experimental setup. n,o) Simulation results comparing the cloaked obstacle (n) with the bare obstacle (o). l-o) Reproduced with permission.^[160] Copyright 2022, Elsevier.



numerical and experimental studies.^[53] A multi-wave metasurface carpet cloak has been designed to conceal objects of arbitrary shapes and sizes under electromagnetic, acoustic, and water waves.^[55]

4.1.5. Water Waveguide Cloak

A broadband waveguide cloak for water waves has been proposed, employing a planar waveguide structure combined with gradient-index metamaterials, demonstrating effective cloaking across a range of water depths and incident wave amplitudes.^[161] The experimental setup is shown in Figure 4h, with the top view of the device in Figure 4i. The internal structure of the gradient-index metamaterial (GIM) is detailed in Figure 4j, and a side view of the assembled cloak is presented in Figure 4k. The cloaking mechanism relies on mode conversion: incoming plane waves are progressively transformed into waveguide-trapped waves (WTWs) within the GIM region and then reconverted into plane waves upon exiting. This approach simplifies water-wave cloaking, as it can be realized by tailoring a graded depth profile along the bottom boundary without additional surface structures. Because the WTWs in the flat section exhibit evanescent behavior, a distinct “cloak region” forms at the center, where wave amplitudes are significantly suppressed. Objects placed within this cloak region experience minimal motion and remain effectively hidden, while the surrounding wave field outside the device is left undisturbed.

4.1.6. Topology Optimization Cloak

Recent advancements in deep learning have provided powerful tools to expedite on-demand design, overcoming the time-consuming, low-efficiency, and experience-dependent limitations of traditional numerical simulations.^[182,183] The integration of artificial intelligence accelerates metamaterial design^[184–187] and promotes intelligent meta-devices.^[188–191] Topology optimization^[192] has gained traction as a flexible numerical method for producing invisibility.^[193] This approach, distinguished by its design flexibility, permits topological changes, such as the creation of holes, throughout the optimization process. The scattering cross-section is the objective function, and the cloak’s composition is optimized iteratively. A water wave cloak is designed and experimentally validated using topology optimization.^[160] Through the inverse design of the water depth distribution, the cloak meets various user-defined scattering criteria without relying on conventional materials. The experimental setup is illustrated schematically in Figure 4m, and a photograph of the fabricated cloak is shown in Figure 4l. This design minimizes the scattering intensity of a cylindrical obstacle

to nearly zero by managing complex scattering events within the cloaking shell, as confirmed by simulation results for the cloaked and bare obstacles (Figure 4n,o). This design minimizes the scattering intensity of a cylindrical obstacle to nearly zero by managing complex scattering events within the cloaking shell. The wavefront remains unchanged when the linear water waves traverse the cloaked object. Therefore, this type of water wave cloak design maximizes the effective use of water waves, presenting potential applications in ocean energy harvesting and harbor protection. The underlying mechanism leverages the interplay of refraction and diffraction induced by variations in bathymetry around the cylinder to neutralize its scattering effect. Even though the proposed cloak is realized at 1.7 Hz, a broadband cloak could theoretically be achieved by optimizing for different target frequencies.

4.2. Water Wave Lenses and Concentrators

Water-wave metamaterials provide powerful mechanisms for manipulating surface waves, particularly through focusing and concentration effects. Two distinct classes of devices have emerged: lenses, which exploit refractive-index engineering to focus or collimate waves, and concentrators, which use transformation-inspired designs to amplify wave energy within designated regions. The historical development is illustrated in Figure 5a.

4.2.1. Water Wave Lenses

When long-wavelength water waves propagate through water containing a bottom-mounted vertical cylinder array, their phase velocity is reduced, resulting in an effective refractive index greater than one. Such structures can therefore be used to achieve the refraction and focusing of water waves, providing a practical means of wave control. This method provides a practical approach for controlling water waves, with potential applications in wave energy harvesting.

The refraction of water waves has been demonstrated using periodic cylinder arrays, and water-wave focusing has been conceptually illustrated.^[38] A biconvex arrangement of rigid cylinders has been engineered to manipulate wave transmission properties, forming a lens that focuses plane liquid surface waves across a wide range of frequencies Figure 5b.^[39] To validate this concept, a pulse of liquid surface waves was generated, and the experimental results (Figure 5c–e) confirm that the lens effectively focuses pulses over a broad frequency spectrum.

The concept of a zero-index refractive medium offers additional opportunities for wave manipulation, attracting significant

Figure 5. Water wave lenses and concentrators. a) Timeline of water wave lenses and concentrators. b) Schematic of water wave focusing using a cylindrical array. c–e) Experimental results showing the focusing of pulsed liquid surface waves. c–e) Reproduced with permission.^[39] Copyright 2009, American Institute of Physics. f) Schematic illustration of surface gravity wave lensing, where incident waves from the left are directed toward a focal point. g) Photograph of the fabricated surface-wave lens composed of smoothly curved seabed ripples. h) Comparison between experimental measurements and direct simulation results of the lens. f–h) Reproduced with permission.^[207] Copyright 2014, American Physical Society. i) Schematic of a water-wave concentrator. j) Simulated amplitude profiles of the concentrator with baffles. k,l) Experimental results of the concentrator at 1.2 Hz. j–l) Reproduced with permission.^[60] Copyright 2018, American Physical Society. m) Experimental setup of the invisibility concentrator. n) Schematic of the invisibility concentrator concept. o) Simulated field patterns comparing a large cylinder alone with the cylinder surrounded by the concentrator. m,o) Reproduced with permission.^[61] Copyright 2020, The Author(s), Published under an exclusive license by AIP Publishing.

attention.^[194–196] In such a medium, waves exhibit infinite phase velocity and wavelength, with no spatial phase change. Consequently, the geometry of the medium's exit surface solely determines the shape of the wavefronts emerging from a zero-index medium, allowing for substantial flexibility in directing the outgoing wavefronts. By covering water with a thick, rigid, immovable plate, the water behaves as a zero-index medium for all wave frequencies. Using concave or rectangular plates, water waves can thus be focused or collimated over a wide frequency range.^[197]

Gradient-index (GRIN) lenses, well established in optics, have also been realized for water waves by gradually varying the filling fraction or draft of cylinder arrays.^[40,198–201] This adaptation represents an analogy to optical GRIN media, as both rely on smoothly varying effective refractive indices to bend and focus waves. However, there are important differences: in optics, the refractive index is varied through material composition, whereas in water-wave systems, it is engineered through bathymetric structuring such as shallow-draft cylinder arrays. Moreover, the effective refractive index in water waves depends on water depth and cylinder draft, and can even fall below unity at critical draft values. For example, a gradient-index medium has been utilized in wave-functional approaches to control linear water surface waves for bending, directional emission, and focusing.^[201] Theoretical and experimental studies of shallow-draft cylinder arrays demonstrate their ability to refract and focus water waves.^[63] GRIN lens designed using shallow-draft cylinders of varying radii achieves a short focal length with high focal intensity, confirming that properly configured low-draft cylinders can bend and amplify water waves across different wavelengths while minimizing surface reflection.

Epsilon-near-zero (ENZ) properties, characterized by vanishing permittivity at specific wavelengths while maintaining finite permeability,^[202–205] can similarly be realized in water waves by surrounding deep-water regions with shallow water, producing a high refractive index contrast. Leveraging metamaterial principles and inherent nonlinearities of water waves enables sub-wavelength harmonic focusing relative to the incident wavelength.^[206] Unlike GRIN lenses, this method allows the creation of topographical lenses through abrupt depth changes based on the ENZ concept. Moreover, seabed corrugations exploiting Bragg resonance properties have been employed to control surface gravity waves, enabling the formation of lenses and mirrors that direct waves toward a designated focal point.^[207] As schematically illustrated in Figure 5f, incident waves from the left are guided by carefully designed seabed ripples to converge at a chosen focal location, which can be positioned upstream or downstream. To realize this concept experimentally, a surface-wave lens was fabricated with smoothly curved seabed ripples (Figure 5g). The lens performance was assessed by comparing experimental measurements with direct numerical simulations (Figure 5h), revealing excellent agreement and demonstrating a focal amplitude approximately three times that of the incident waves, confirming the effectiveness of the design in concentrating surface gravity waves.

These approaches demonstrate that water-wave lenses—whether based on periodic arrays, zero-index concepts, GRIN media, or seabed structuring—offer versatile and effective strategies for wave focusing and control.

4.2.2. Water Wave Concentrators

In contrast to lenses, which focus waves via refraction and geometry, water-wave concentrators aim to amplify wave fields internally while leaving external fields largely unaffected. These designs are rooted in transformation optics and exploit inhomogeneous depth profiles or refractive indices to guide energy into confined regions.^[208,209]

One approach uses gradient depth profiles to minimize boundary scattering, resulting in efficient wave concentration.^[60] The design principle is schematically illustrated in Figure 5i. Simulated amplitude profiles with baffles (Figure 5j) illustrate effective wave energy concentration, while experimental measurements at 1.2 Hz (Figure 5k–l) confirm substantial amplitude amplification, demonstrating the concentrator's ability to efficiently focus water-wave energy.

Extending cloaking principles, a water-wave invisibility concentrator has been realized using multiple truncated cylinders arranged concentrically.^[61] The experimental setup is shown in Figure 5m, while the working principle of the concentrator is schematically illustrated in Figure 5n, where incident waves are guided around the central object to suppress scattering. Simulated field patterns (Figure 5o) compare a large cylinder alone with the cylinder surrounded by the concentrator, demonstrating that the device effectively focuses scattered waves internally. Experimental validation using a float placed inside confirms amplified motion, highlighting the concentrator's ability to combine wave guidance with cloaking functionality.

Concentrator concepts have also been applied to energy harvesting. By dividing the water domain into three zones with distinct properties, a highly efficient device was engineered to capture the kinetic energy of slow-moving water flows.^[210] These results highlight the practical utility of concentrators for renewable energy applications.

Overall, water-wave concentrators complement lenses by offering a distinct form of wave manipulation: rather than merely focusing external wavefronts, they amplify energy within a region while maintaining external field invisibility. This duality underscores the richness of metamaterial-inspired strategies for wave control.

4.3. Superscatterer for Water Waves

Superscattering, where the scattering cross-section of a subwavelength object exceeds the single-channel limit, offers promising opportunities for enhanced energy harvesting, sensing, bioimaging, and communication.^[211–214] Extending the principles of superscattering from optics to hydrodynamics, this phenomenon has been experimentally realized for water waves.^[65] To construct a subwavelength superscatterer, multiple concentric cylinders with varying water depths were precisely designed. By carefully aligning the scattering peaks from different angular momentum channels, the total scattering cross-section can exceed the single-channel threshold.

The experimental setup for the water-wave superscatterer is illustrated in Figure 6a, showing how incident waves interact with the structured scatterer. The superscatterer was fabricated using advanced 3D printing techniques and tested in controlled

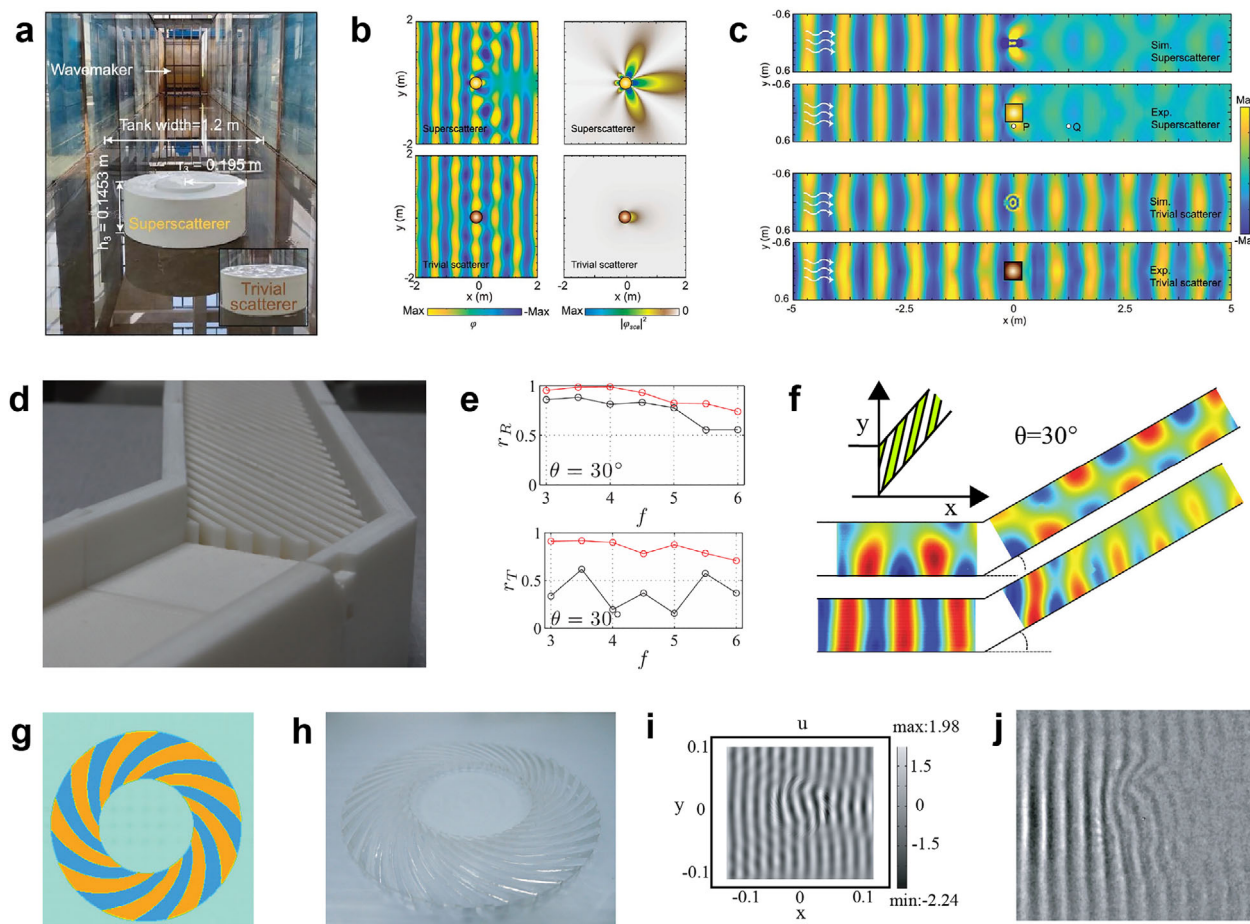


Figure 6. Water wave superscatterer, shifter, and rotators. a) Schematic of the experimental setup for the water-wave superscatterer. b) Simulation results showing total field patterns for both the superscatterer and the trivial scatterer, along with their respective scattering energies. c) Comparison between measured and simulated field patterns of the superscatterer and the trivial scatterer. a–c) Reproduced with permission.^[65] Copyright 2022, The Author(s), Published by Oxford University Press on behalf of China Science Publishing & Media Ltd. d) Design of a water-wave metamaterial shifter. e) Mode 0 contribution to the total field—reflection ($X < 0$, top) and transmission ($X > 0$, bottom) versus frequency—red curves represent the metamaterial waveguide and black curves the reference waveguide. f) Measured surface elevation fields in the reference (top) and metamaterial waveguide (bottom) at $f = 4$ Hz. d–f) Reproduced with permission.^[11] Copyright 2013, American Physical Society. g, h) Design and fabricated sample of a metamaterial rotator. i, j) Simulation and experimental results of the metamaterial rotator at $f = 5$ Hz. g–j) Reproduced with permission.^[64] Copyright 2009, The Institute of Physics.

finite water tanks. Numerical simulations (Figure 6b) compare the total field patterns of the superscatterer and a trivial scatterer, along with their respective scattering energies, highlighting the enhanced scattering effect of the superscatterer. Experimental measurements (Figure 6c) show strong agreement with the simulations, confirming the superscatterer's ability to significantly amplify scattered wave fields compared with a conventional scatterer. A plastic boat placed in proximity exhibits amplified or diminished vertical motion corresponding to the local wave amplitude, highlighting the device's influence on surrounding water waves.

This mechanism not only demonstrates the ability to strongly manipulate water wave propagation but also suggests practical applications: the amplified fluctuations near the superscatterer can be harnessed for energy harvesting, while the large forward shadow could be strategically employed for harbor protection.

4.4. Water Wave Shifters and Rotators

Metamaterials for water waves have facilitated the development of devices capable of redirecting and reorienting wave propagation. Among them, wave shifters and rotators represent two fundamental classes of transformation-based designs that manipulate wave trajectories while leaving the external field largely undisturbed.

4.4.1. Water Wave Shifters

Water-wave metamaterial shifters rely on transformation media to guide waves along tailored trajectories without reflection. A representative demonstration is the realization of a reflectionless bent waveguide, achieved using an anisotropic medium composed of bathymetric structures with subwavelength layers.

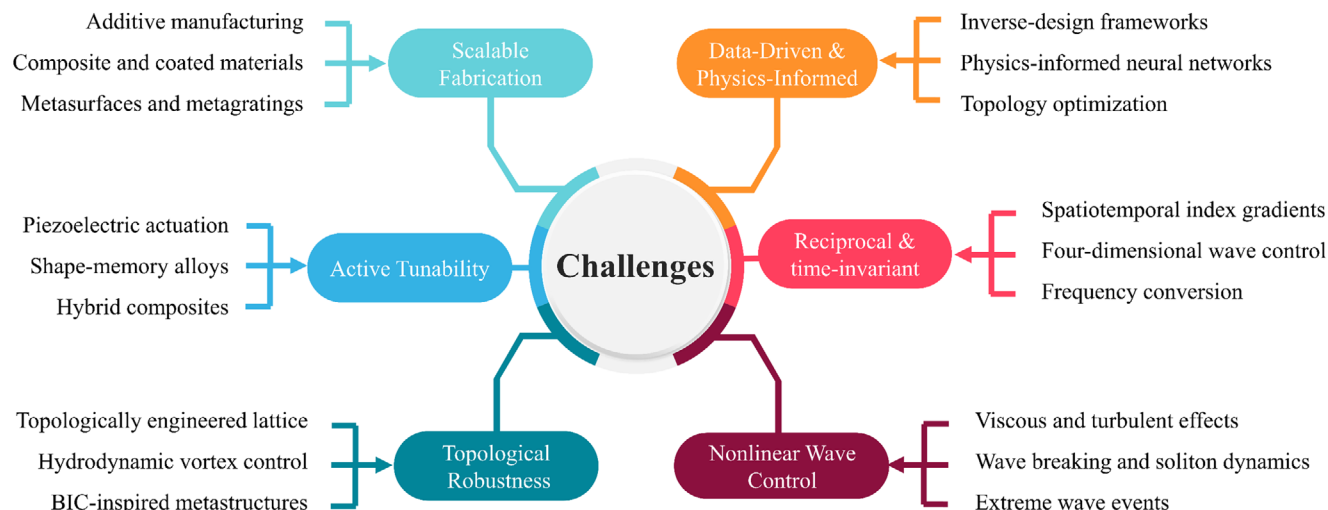


Figure 7. Roadmap for water wave metamaterials.

This configuration effectively bends and redirects wave propagation while preserving transmission efficiency.^[11] The design of such a metamaterial shifter is illustrated in Figure 6d, where the structural arrangement governs wave control. Figure 6e shows the mode 0 contribution to the total field, presenting reflection ($X < 0$, top) and transmission ($X > 0$, bottom) spectra; red curves correspond to the metamaterial waveguide, while black curves represent the reference. Experimental surface elevation measurements at $f = 4$ Hz (Figure 6f) further confirm the shifter's effectiveness, demonstrating clear differences between the reference waveguide (top) and the metamaterial waveguide (bottom).

4.4.2. Water Wave Rotators

In contrast to shifters, metamaterial rotators manipulate the coordinate system itself to redirect the internal propagation direction of waves. The concept, initially demonstrated for electromagnetic waves,^[215,216] acoustic waves,^[217] heat transfer,^[218–220] and fluid flow,^[221] has been adapted to water waves through layered bathymetric designs. Inspired by electromagnetic wave rotators, a spirally arranged layered structure has been developed to impose anisotropic depth, thereby rotating water-wave energy flow relative to the surrounding field.^[64] This design functions analogously to an invisibility cloak: while the internal field is rotated, the external field remains unaffected, rendering the rotated region “invisible” to detection. Figure 6g–h shows the conceptual and fabricated structures of the water-wave rotator, while Figure 6i–j presents simulation and experimental results at $f = 5$ Hz, confirming the ability of the device to achieve controlled rotation of wave propagation.

5. Conclusions and Future Directions

The research on metamaterials for water waves has opened up promising opportunities in coastal engineering, renewable energy harvesting, underwater acoustics, and environmental mon-

itoring. Nevertheless, the path toward practical deployment requires addressing several outstanding challenges related to scalability, fabrication, nonlinearity, and tunability (Figure 7).

Most existing demonstrations remain restricted to laboratory-scale prototypes with simplified geometries. Scaling these concepts to realistic oceanic environments requires cost-effective and robust fabrication strategies. Promising avenues include 3D printing of graded materials, large-area molding of seabed structures, and modular assembly of metamaterial units. For instance, 3D-printed graded acoustic metamaterials have already been scaled to meter-sized devices for underwater applications,^[222] providing a model for upscaling water-wave metamaterials. Equally important is ensuring the long-term durability of structures under harsh marine conditions, including salinity, biofouling, and erosion.

Advances in computational approaches, particularly deep learning and inverse design, are set to significantly accelerate the development of metamaterials.^[223–230] Deep learning has achieved remarkable success across a variety of domains, including science,^[231–233] medicine,^[234–236] and object detection.^[237,238] A viable workflow can be envisioned: high-fidelity simulations generate datasets for surrogate models, which in turn enable deep learning frameworks to rapidly predict wave–structure interactions without costly solvers. Building on this capability, inverse design algorithms can identify optimal geometries for functionalities such as cloaking or focusing. Closed-loop adaptive control, integrating AI with real-time sensing, could ultimately yield intelligent water-wave metamaterials capable of dynamic responses. This workflow, already demonstrated in electromagnetic metasurfaces,^[226,239] can be translated to water-wave systems, as evidenced by early studies.^[160]

Physics-informed neural networks (PINNs)^[240] are poised to provide a promising framework for advancing the design and analysis of water-wave metamaterials. By embedding the governing hydrodynamic equations, such as the Navier–Stokes equations, directly into the learning architecture, PINNs enable the network to learn effectively from limited or noisy experimental data while preserving physical consistency. This hybrid approach

allows the model to capture nonlinear dispersive effects, viscous dissipation, and complex boundary interactions that are often difficult for purely data-driven methods to represent. In addition, PINNs may be applied to inverse design, real-time system identification, and adaptive control of water-wave metadefices, including under dynamically varying marine conditions. Overall, PINNs have the potential to significantly expand the modeling and optimization paradigm for future intelligent and adaptive water-wave metamaterials.

Incorporating tunable materials represents another compelling avenue. For example, piezoelectrics have enabled tunable band gaps in acoustic metamaterials,^[241] while shape-memory alloys have been employed in reconfigurable mechanical metamaterials.^[242–246] Embedding such materials into water-wave devices could allow real-time adaptation to changing sea states, enabling breakwaters to dynamically mitigate storm surges or optimize energy capture. Inspiration from nature—such as marine organism structures—offers further opportunities for biomimetic designs that are both efficient and resilient in diverse environments.

Recent progress in space–time metasurfaces also highlights opportunities for water-wave systems. Time-reversed waves have been demonstrated in acoustic,^[247] elastic,^[248] electromagnetic,^[249] and water waves.^[250–252] Space–time modulation enables simultaneous control over spatial and temporal frequencies, opening a frontier in four-dimensional wave manipulation.^[253–256] Despite this progress in electromagnetics, the exploration of space–time water-wave metamaterials remains limited, making it a highly promising future direction.

Beyond metamaterials, artificial structures inspired by topological wave physics are emerging as another frontier.^[257,258] Examples include topological bound states in the continuum, topologically protected edge modes, vortices, and skyrmions, which could enable robust, defect-immune water-wave transport.^[259–272] Hydrodynamic moiré superlattices^[273] further expand the design landscape. Moreover, analogous to Cherenkov radiation in electromagnetics, engineered water-wave crystals can tailor forward and backward wakes with tunable radiation angles, enabling applications in wake stealth, navigation, and offshore fluid dynamics.^[274] Recent demonstrations of valley vortex states in water-wave crystals^[275] establish water waves as a powerful analog platform for exploring topological phenomena with direct relevance to marine engineering and ocean energy.

The limitation of current models is the treatment of water as an inviscid, incompressible medium, neglecting viscous and nonlinear effects. Real ocean environments inherently involve wave breaking, turbulence, and viscosity. The emerging concept of hydrodynamic metamaterials^[276–278] provides a framework to incorporate viscous effects and achieve realistic manipulation of fluid flows. Nonlinear wave control—such as harmonic generation, modulational instability, and soliton steering—remains underexplored yet holds potential for transformative applications.

As research progresses, water-wave metamaterials are poised to transform our interaction with water waves, offering sustainable and efficient solutions to pressing societal challenges. Continued innovation and cross-disciplinary collaboration will be essential to realize their full potential, ushering in a new era of wave manipulation and materials science.

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Conflict of Interest

The authors declare no conflict of interest.

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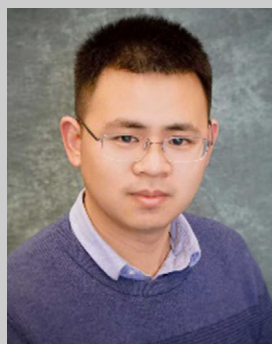
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